UNIT-3

Electrochemistry and Applications.

Introduction: Electrochemistry is the study of process involving the interconvention of electrical energy and chemical energy. In brief it deals with the chemical applications of electricity.

A cell is a single arraingrent of two electrodes and an electrolyte. A cell may be used for two purposes.

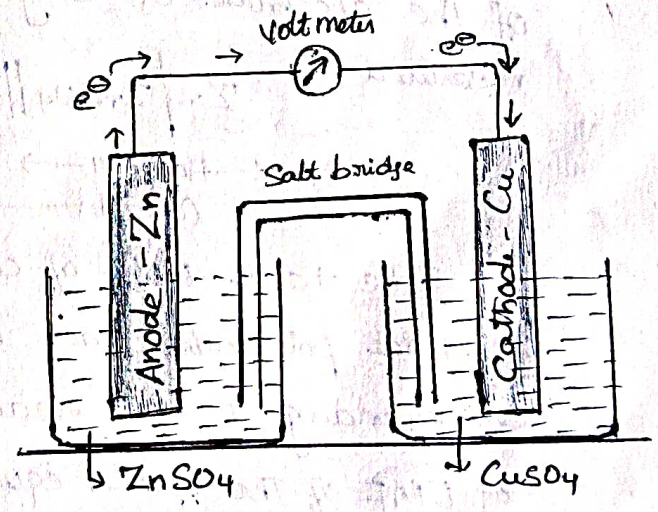
* → To convert chemical energy into electrical energy. Such type of cells are called "Electrochemical cells".
  Ex: Galvanic cell or Voltaic cell, Drycell.
* → To convert electrical energy into chemical energy such type of cells are called "Electrolytic cells".
  Ex: These are used for the purification of the metals.

Electro Chemical Cell:→

Electrochemical cell which converts chemical energy to electrical energy. Ex: Galvanic cell.

Galvanic cell:- Galvanic cell also called as a voltaic cell. In which electrical energy or current is generated by a spontaneous Redox reactions.

$$Zn + Cu^{+2} \longrightarrow Zn^{+2} + Cu.$$



A bar of Zinc metal (anode) is passed in $ZnSO_4$ solution (electrolyte), In the left container. A bar of Cu (cathode) is placed in $CuSO_4$ solution. in right container.

The Zn & Cu electrodes are joined by Cu wire. a salt bridge containing K_2SO_4 solution, Inter converts the solution is the two containers.

Oxidation reaction: $Zn \longrightarrow Zn^{+2} + 2e^{\ominus}$

Reduction reaction $Cu^{+2} + 2e^{\ominus} \longrightarrow Cu$

$Zn + Cu^{+2} \longrightarrow Zn^{+2} + Cu$ (Redox)

[2] When the cell is setup Zn dissolves in the anode sol^{n} to form Zn^{+2} ions then electrons flow from Zn electrode through wire to Cu electrode. The Cu^{+2} ions in the cathode solution gain the electrons and converts to Cu atoms. In this process the salt bridge can equalise the ions through migration proces and which gives continuous supply of current.

The total cell according to the convention is represented as

$Zn/ZnSO_4 \| CuSO_4/Cu$ $\quad E = 1.1$ volt.

Cell potential (EMF of the cell):- The voltage of potential difference b/w the electrodes of an electrochemical cell is called Electro motive force (EMF) of the cell.

Single electrode potential:→ The potential difference between electrode and its electrolyte in which it is dipped is called potential of the electrode or single electrode potential. It can't be directly measured.

$Zn/ZnSO_4 \| CuSO_4/Cu$ $\quad E_{(cell)} = E_{(anode)} + E_{(cathode)}$.

← Half cell → (anode) ← Half cell → (cathode).

The total EMF of cell is equal to the sum of the single electrode potentials. Each electrode is affixed with a symbol corresponding to the reaction that takes place near the electrode

→ EMF of the cell is equal to the sum of oxid^{n} potential and reduction potential also expressed.

$$E_{(cell)} = E_{Right} - E_{left}.$$

Generally electrode potential refers to reduction potential.

Ex: $E_{(R)} = Cu^{+2}/Cu$ is 0.337 V

$E_{(L)} = Zn/Zn^{+2}$ is -0.763 V

$E_{cell} = E_R - E_L$

$= 0.337 - (-0.763)$

$= 0.337 + 0.763$

$= 1.1$ V

Electrochemical Series:

The electrochemical series is a list of elements arranged based on their standard electrode potentials. It helps predict the direction of redox reactions.

For the example, metals higher in the series are more likely to undergo oxidation, while those lower tend to be reduced.

Standard Oxidation Potential Values at 25°C.

Electrode	Halfcell rxn	E° Volts	
$Li\|Li^+$	$Li \longrightarrow Li^+ + e^-$	+3.045	Active anode.
$K\|K^+$	$K \longrightarrow K^+ + e^-$	+2.925	
$Mg^{+2}\|Mg$	$Mg \longrightarrow Mg^{+2} + 2e^-$	+2.37	
$Zn\|Zn^{+2}$	$Zn \longrightarrow Zn^{+2} + 2e^-$	+0.763	
$Fe\|Fe^{+2}$	$Fe \longrightarrow Fe^{+2} + 2e^-$	+0.440	
$Pb\|Pb^{+2}$	$Pb \longrightarrow Pb^{+2} + 2e^-$	+0.126	
$Pt\|H_2(g)\|H^+(1m)$	$H_2 \longrightarrow 2H^+ + 2e^-$	0.000	
$Cu\|Cu^{+2}$	$Cu \longrightarrow Cu^{+2} + 2e^-$	−0.337	Noble Cathode
$Ag\|Ag^{+2}$	$Ag \longrightarrow Ag^{+2} + 2e^-$	−0.799	
$Au\|Au^{+3}$	$Au \longrightarrow Au^{+3} + 3e^-$	−1.50	

Applications of Electrochemical Series

1. To compare the relative oxidizing and reducing powers
2. To calculate the relative activities of metals
3. To calculate the standard EMF of any electrochemical cell.
4. To predict whether a redox reaction is spontaneous.
5. To study the reducing nature of metals.
6. Corrosion protection, Battery design, and Corrosion testing.

Electrochemical Sensors:-

Electrochemical sensors are devices that detect and measure chemical substances by converting the chemical information into an electrical signals. They typically consists of electrodes immersed in a solution containing the target analyte. The interaction b/w the analyte and electrodes generates a measurable electrical response. [allowing for the quantification] such as voltage or current.

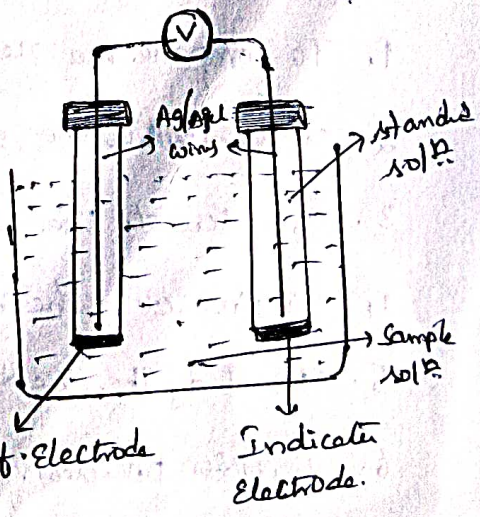
Types:- * Amperometric sensors
* potentiometric sensors
* Conductometric sensors.

These sensors find applications in various fields, from environmental factors, like detecting gases, pollutants, and biomolecules, etc.

potentiometric Sensors:-

potentiometric sensors measure potential difference b/w electrodes in response to a chemical reaction. They depend on an electrochemical cell with one analyte-sensitive electrode. Interaction with the analyte generates a potential difference, measured against a reference electrode. This output informs about target substance concentration. widely applied in environmental monitoring and industrial control. Commonly used in pH meters and ion-selective electrodes.

potentiometric sensors are sensitive to the Nernst eqn which describes the relationship b/w the Con. of ions and the electrode potential. By measuring the voltage, the sensor can provide information about the analyte Concentration.



These sensors convert chemical reactions into voltage changes, enabling the precise measurement of analyte concentrations in solutions.

Potentiometric Titrations

Potentiometry:→ potentiometry is a technique in analytical chemistry that measures the voltage or potential difference b/w two electrodes in a solution.

This method is commonly used to determine the concentration of ions or molecules in a sample by exploiting the relationship b/w electrochemical potential and analyte concentration.

Potentiometric Titration:-

1) potentiometric titration is a technique similar to direct titration, but in this, no indicator reagent is used.
2) In potentiometric titrations, a cell is used with reference electrode (Ex: Hydrogen electrode, Ag-cl electrode, Calomel electrode), salt bridge, analyte and an indicator electrode (such as Glass or metal ion electrode).
3) When the pair of electrodes are placed in the sample solution or analyte, it shows the potential diff. b/w two electrodes by the addition of the titrant or by the change in the conc. of ions.

Redox Titration:→

→ In this process, to determine the % of Fe^{+2} in an unknown sample mixture through redox titration using a standard $K_2Cr_2O_7$ solution.

→ Prepare the Fe^{+2} and $K_2Cr_2O_7$ solutions as per concentrations. Fe^{+2} sol^{n} prepared in acedic medium, for redox reactions.

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$$

$$Fe^{+2} \longrightarrow Fe^{+3} + e^-$$

$$Cr_2O_7^{2-} + 6Fe^{+3} + 14H^+ \longrightarrow 2Cr^{3+} + 6Fe^{+3} + 7H_2O$$

→ Calibrate the potentiometer and ensure its functioning properly.

→ place the two electrodes (Ref. electrode and indicate electrode) in Fe^{+2} solution through potentiometer.

[6]

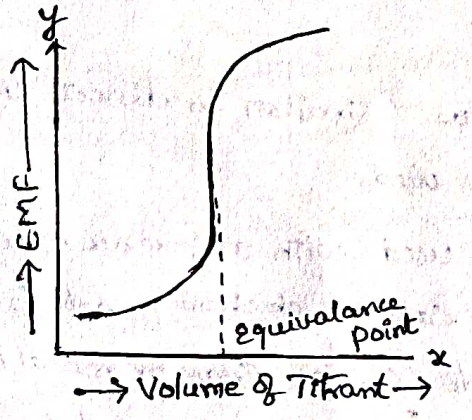
→ Observe the initial EMF value of the sample solution

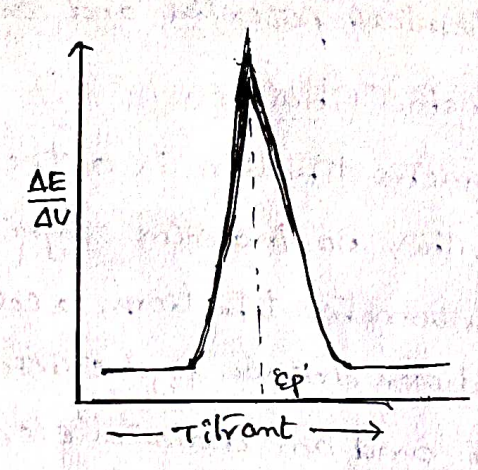
→ Take $K_2Cr_2O_7$ sol^n in burette, now add slowly to the Fe^{+2} sol^n

→ Record the potential changes continuously using potentiometer, as titration progress.

→ Identify the equivalence point (end point) by monitoring a sudden change in potential.

→





→ Note the volume of added $K_2Cr_2O_7$ at the endpoint

→ Draw graph b/w EMF values and Volumes of titrant ($K_2Cr_2O_7$) with help of potential values.

Conductivity:- conductivity is a measure of a solution's ability to conduct electricity.

Conductometric titrations:

Conductometric titration is the volumetric titration (analysis) based upon the measurement of the conductance during the course of titration.

The no. of free ions, charge of the ions, and mobility of ions affects the conductance of an electrolyte solution. When one electrolyte is added to another electrolyte, the change in no. of free ions causes a change in the conductance.

These Conductometric titrations done by using conductivity cell. and Conductivity meter. And these titration main importance is without using indicater to find the end point of the titration.

I. Conductometric titration between Strong acid Vs Strong base:

i) In this process, Strong acid takes in beaker and Strong base (NaOH) take in burette. And conductivity cell place in acid, note the initial value.

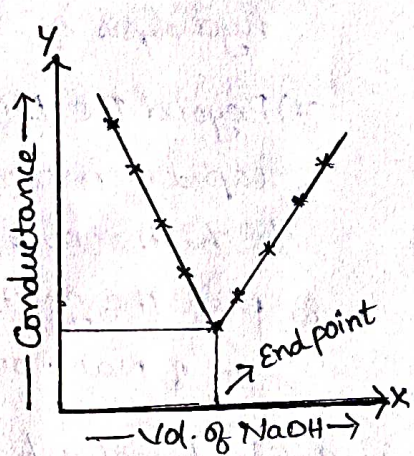
ii) When a Strong acid (HCl) is titrated against a Strong base (NaOH), before NaOH solution is added from the burette, the acid solution has high conductivity due to the high mobility ions (H^+).

iii) When NaOH is added to the acid, the conductivity of acid sol^n decreases due to neutralization of high mobile H^+ ions of the acid with $OH^\ominus$ ions of the base.

$$H\overset{+}{C}l^- + \overset{\oplus}{N}aO\overset{\ominus}{H} \longrightarrow \overset{\oplus}{N}a\overset{\ominus}{C}l + H_2O$$

iv) Thus the conductance of the solution continues to decrease until the equivalent point is reached.

v) Further addition of NaOH solution will increase the conductance by highly mobile hydroxyl ($OH^\ominus$) ions.



vi) The point of intersection of the graph plotted by Conductance (Y-axis) of the solution against volume of alkali (NaOH). (x-axis), added corresponds the end point of the titration.

II. Conductometric titration between weak acid (CH_3COOH) Vs Strong base (NaOH):

1. In this process, Weak acid takes in beaker, and place a conductivity cell, now note initial value of acid. And fill the burette with Strong base.

2. When weak acid (CH_3COOH) is titrated against a strong base (NaOH), at the begining of the titration, the solution contains CH_3COOH, so conductivity is relatively low. Because there is low mobility ions.

3. When NaOH is added to acetic acid, to form acetate ions and Sodium acetate and water.

[8]

$$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$$

acetic acid Sodium hydroxide Sodium acetate water

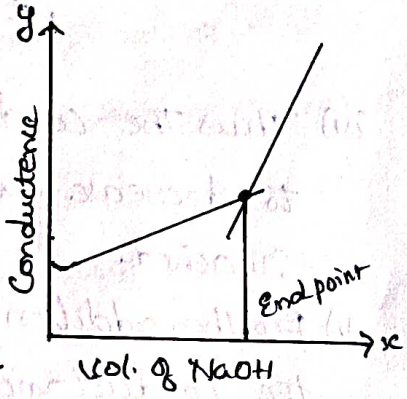
4) Since the acid is weak, its conductivity is relatively low. on the addition of base. there is decreased in conductance not only due to the replacement of H^+ by Na^+ but also suppresses the dissociation of acetic acid due to common ion acetate.

5) But very soon, the conductance increases on adding NaOH. and form CH_3COONa and H_2O.

6) This increase in conductance continues raise up to the equivalence point. The graph near the equivalence point is curved due to the hydrolysis of salt CH_3COONa.

7) Beyond the Equivalence point, Conductance rapidly increases with addition of NaOH due to high mobile ions like OH^-

8) plot the graph b/w Conductance on y-axis, and Vol. of NaOH on x-axis. Find the equivalence point.

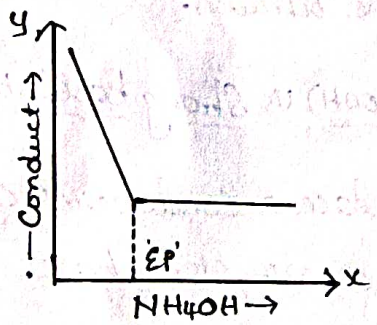


III Conductometric titration b/w SA(HCl) – WB(NH_4OH) :→

* Initially the conductance is high and then it decreases due to the replacement of H^+. But after the End point has been reached. The graph becomes almost horizontal. Since the excess NH_4OH

$$HCl + NH_4OH \longrightarrow NH_4Cl + H_2O.$$

is not appreciably ionised in presence of NH_4Cl.

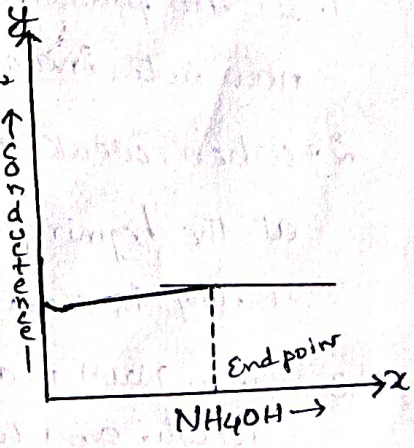


IV Conductometric titration b/w WA(CH_3COOH) – WB(NH_4OH) :→

* The nature of curve before the equivalence point is similar to the curve obtained by titrating CH_3COOH(WA) against weak base (NH_4OH).

$$CH_3COOH + NH_4OH \longrightarrow CH_3COONH_4 + H_2O.$$

→ after end point, Conductivity remains same, as the WB which is being added is freebly ionised & therefore is not much Conductance



* * * *

Batteries:- Battery is a device, which converts chemical energy into electrical energy is called Battery or cell, which is an electrochemical cell or often several electrochemical cells connected in series that can be used as a source of direct current at a constant voltages. Batteries are commercial electrochemical cells. Redox reactions takes place.

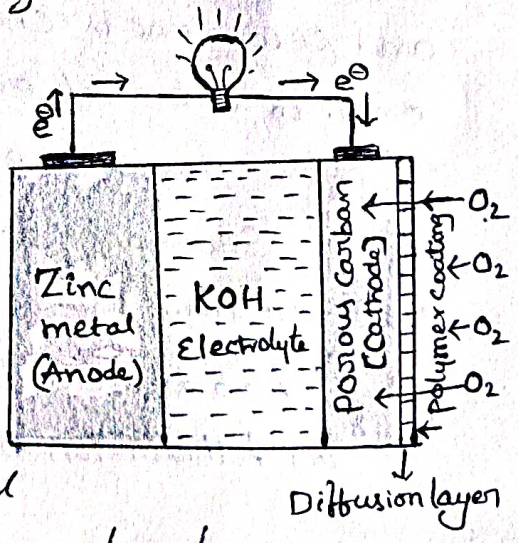
Batteries are classifies as i) primary cells (primary batteries)
ii) Secondary cells (Sec. batteries).
iii) Fuel cells.

i) Primary Batteries or primary Cells:→

A primary battery acts as an electrochemical cell. These are non-rechargeble type of batteries that produce electric power through chemical reactions (Convertion of chemical energy to electrical energy). In these cells, a single discharge cycle is observed. once the chemical in battery are used up, the battery will be dead, and can't be recharged for additional use.

Ex:- Dry cell or Leclanche cell, Zn-air cells, Mercury batt, etc.

Zn-Air battery:→ Zinc-air battery's are a type of ele.chemical battery that used Zn as the anode and Oxygen from air as the cathode. (a porous carban rod serves as cathode). These batteries generate electrical power through a chemical reaction, b/w Zn and O_2. porous carban rod is coated with polymer coating and gas diffusion layer, which are used to absorbs and diffuse the oxygen from surroundings and avoid any impurities and moisture.



Zn electrode undergoes oxidation, and, releasing electrons that flows external circut, producing electricity.

$Zn/KOH/O_2$

(10) Oxygen from atmosphere diffuses into the porous electrode (at cathode), and gains electrons, and reduction process takes place. Cell reactions given below. And the EMF of cell is - 1.4 V. to 1.6 V

$$KOH \longrightarrow K^+ + OH^-$$

at Anode:

$$Zn + 2OH^- \longrightarrow ZnO + H_2O + 2e^- \text{ (Oxi)}$$

at cathode: $1/2 O_2 + 2e^- + H_2O \longrightarrow 2 \cdot OH^-$ (Reduction).

Net rxn: $Zn + 1/2 O_2 \longrightarrow ZnO$.

Applications :-

1. Zinc-air batteries are commonly used in hearing aids due to their compact size, light weight.
2. used in medical devices.
3. Military applications.
4. used in portable electronics like watches, pagers, calculators.
5. used in Remote communications, voice transmitters, etc..

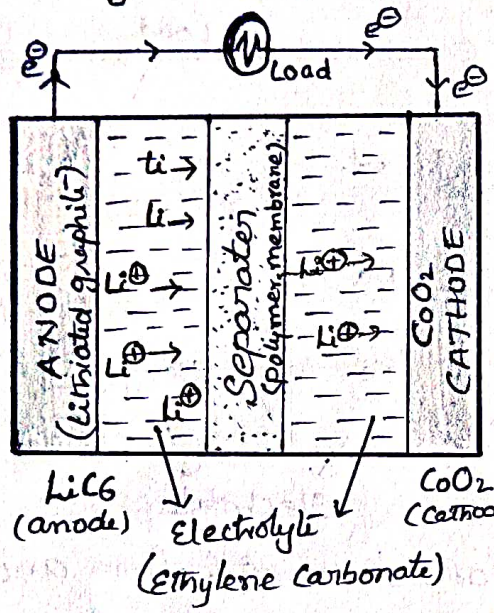
ii) Secondary Cells :→ A Secondary battery acts as both, Electrochemical cell and electrolytic cell. These are rechargable type of batteries. In this batteries two cycles are observed, one is discharge cycle (Conversion of chemical energy to Electrical ener) and second one is charge (or) recharge cycle (Convertion of electrical ener. to chemical energy).

These are electrically charged and discharged multiple times. Secondary cells are designed for repeated use. These are used electronic devices, e-vehicles etc.

Ex :- i) Lead-acid batteries
ii) Ni-metal hybride cells
iii) Lithium-ion batteries
iv) Na-ion batteries.

Lithium-ion batteries :→ (LIB-batteries)

→ Lithium ion batteries is rechargeble energy storage device that utilizes Li ions to move between positive and negative electrodes during charging and discharging. These battery produce 1.5 to 3V.



→ In this cell Lithiated graphite acts as anode, and CoO_2 (Cobalt oxide) acts as cathode, Ethylene carbonate acts as electrolyte.

→ The separator (polymer membrane) is a thin membrane, that separates, anode and the cathode.

Working :→

→ Lithium is stored in anode and cathode. The Electrolyte carries the positive charged Lithium ions from the anode to cathode, and vice versa, through a separator

Discharge process:- During the discharging process the anode releases Li^+ ion to the cathode and generates an electron flow from one side to other side through circuit / load, during this process electrical current generated. And Li^+ ions deposit at cathode.

$$LiC_6 \rightleftharpoons Li^+ + C_6 + e^\ominus$$

Charging process:- During the charge/recharge process, the Li ions coated cathode (CoO_2) gets electrons to form $LiCoO_2$ & the Li ions again flow to reach anode through electrolyte, separater.

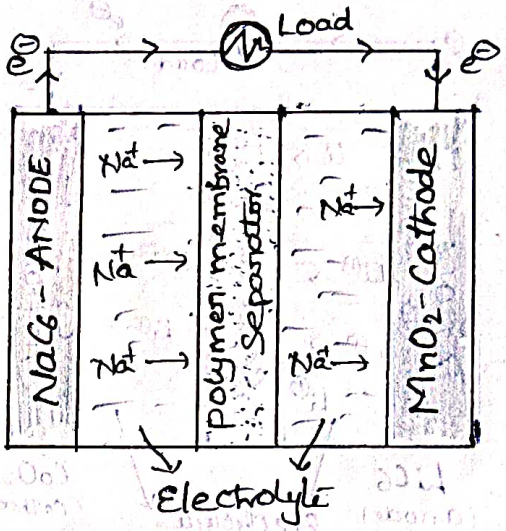
$$CoO_2 + Li^+ + e^\ominus \rightleftharpoons LiCoO_2$$

Applications:-

1) used in Laptop, mobiles.
2) used in Electric cars, buses, etc
3) Health instruments - pacemakers,
4) In UPS power backup systems
5) portable power packs,
6) Digital Cameras, etc.

Sodium-ion batteries:

* Sodium-ion battery (NIB or SIB) is a type of rechargeable battery that uses Na^+ ions as its charge carrier.
* The working principle and cell construction are similar to those of Li-ion battery, but it replaces Li ions with Na as the anodic material.
* Sodium-ion batteries have more charge cycles than Li-ion and develops a potential of 3.0 V.



Discharge process:

During the discharge, Sodium ions dissociated from anodic material and migrate towards the cathode through electrolyte and a separator.

$$NaC_6 \rightleftharpoons Na^+ + C_6 + e^\ominus$$

At the same time, the electrons travel through the external circuit

Recharge process: During recharge/charging process, Na from cathodic material is ionized and move towards the anode with help of electrons.

$$Na^+ + MnO_2 + e^\ominus \rightleftharpoons NaMnO_2$$

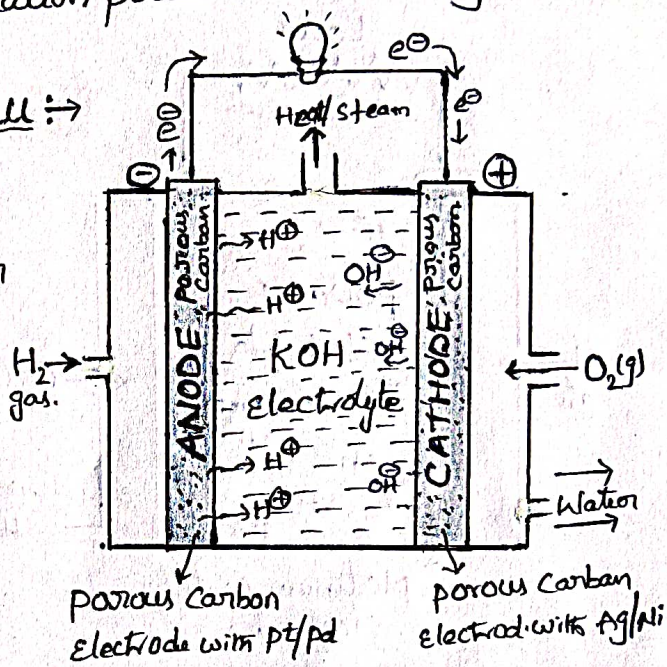
Applications:

1. used in ups applications in the telecommunications sector.
2. used in numerous stationary energy storage applications
3. used in large-scale storage of renewable energy.
4. used in grid-level applications.
5. power backup. etc.

FUEL CELLS:- A fuel cell is an electrochemical device, that converts the chemical energy from a fuel directly into electri-city through an electrochemical reaction (Redox reactions). usually the hydrogen, hydrocarbon, CH_3OH, etc. are reacts with O_2 in this device then produce electricity continuously.

$$\text{Fuel} + O_2 \longrightarrow \text{Oxidation product} + \text{Electricity}$$

1. Hydrogen-oxygen (H_2-O_2) fuel cell :-



a) In H_2-O_2 fuel cell hydrogen (H_2) gas is diffuced through a porous carbon electrode, which is coated with platinum (or) palladium catalyst. These eclectrode acts as anode.

b) Oxygen (O_2) gas is diffuced through a porous carbon electrode, which is coated with Silver (Ag) [or] Nickel (Ni) These electrode acts as cathode.

c) The electrodes seperated by KOH (electrolyte).

d) The diffuced H_2 gas splits into H^+ ions and electrons, at anode, and electrons travel through the external circuit to cathode, and generates electricity (oxidation proces takes place).

e) O_2 gas from cathode gets electrons to do reduction process.

f) EMF of the H_2-O_2 fuel cell is 1.23V. And these are ecofriendly.

$$KOH \longrightarrow K^+ + OH^-$$

- at anode: $2H_2 + 4OH^- \longrightarrow 4H_2O + 4e^-$ (Oxidation)
- at cathode: $O_2 + 2H_2O + 4e^- \longrightarrow 4OH^-$ (Reduction)

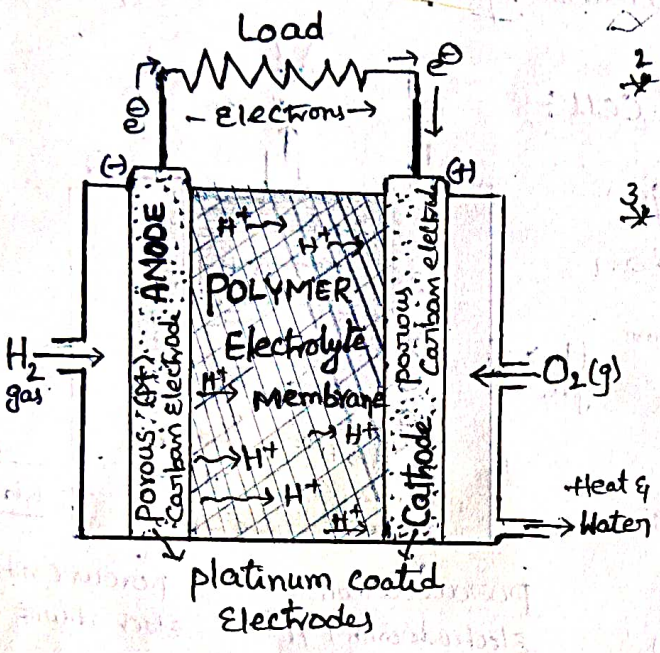
$$2H_2 + O_2 \longrightarrow 2H_2O \quad \text{(Redox)}.$$

Applications:- i) used in space vehicls, Submerine energy sources,
ii) used in militery vehicls.
iii) used Supply electricity on earth.

(14)

PEMFC [Polymer Electrolyte Membrane Fuel Cell] :→

* In this fuel cell polymer membrane (such as polystyrene sulphonic acid) acts as electrolyte. These cells is also called as "proton Exchange membrane fuel cell (PEMFC)."



1* PEM cell constructed by porous carbon electrodes (those are coated by 'Pt').

2* Electrodes i.e anode and cathodes are separated by polymer membrane.

3* H_2 gas diffuses to anode side they splits into H^+ and electrons, at the same time oxygen gas enter from cathode side, they combine to electrons, and H^+, generating water and heat.

4* At this time Redox reactions takes place (at anode Oxidation and at cathode Reduction).

4* These cells operates at low temperature.

5* These cells are low weight and volume, when compared to other fuel cells, the EMF of these cells is 1.23 V. These are eco-friendly cells.

6* The electrons travel through external circuit from anode to cathode.

7* The PEM, which allows rapidly hydrogen ions(H^+) from anode to cathode, and from water.

Cell reactions:-

at Anode: $H_2 \longrightarrow 2H^+ + 2e^\ominus$ (Oxidation)

at cathode: $\frac{1}{2}O_2 + 2H^+ + 2e^\ominus \longrightarrow H_2O$ (Reduction)

Net : $H_2 + \frac{1}{2}O_2 \longrightarrow H_2O + \text{Energy}$ (Redox).

Applications :→

a) used in automobile powering systems.

b) used for backup power for commercial industries, residential buildings, & remote areas. c) used portable power systems.

—— ** All the Best ** ——